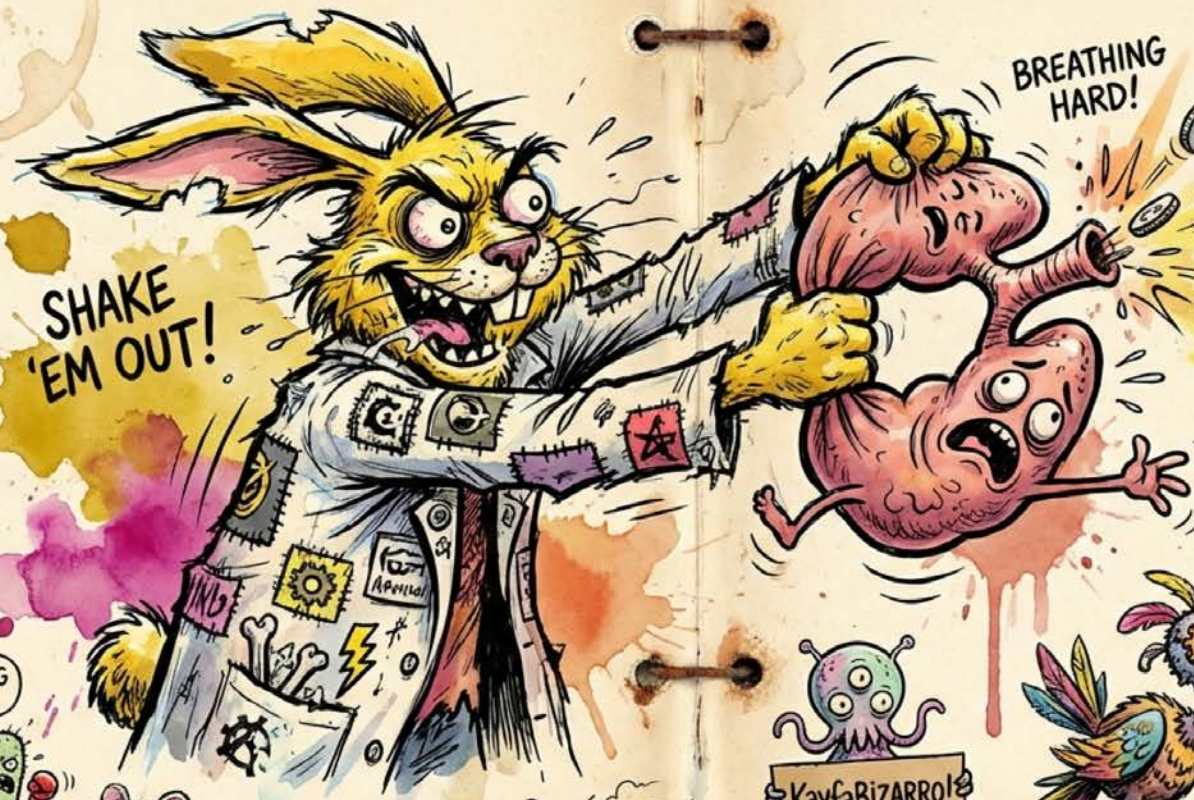
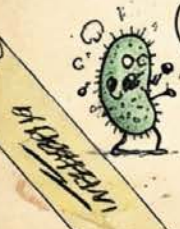
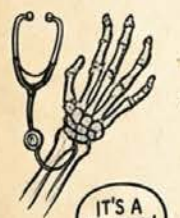
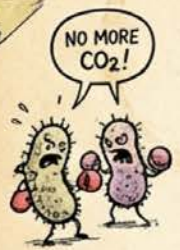


MAYFABIZARRO!

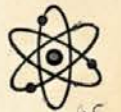
# MEDKAYFAB: PNEUMOLOGY WRESTLING!

Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kards for Narrative Medical Learning Games.

MASK ZIANE



FIGHT THE CO<sub>2</sub> MONSTER!



MAYFABIZARRO!



# ALFIE THE ALVEOLUS (THE FRAGILE BUBBLE)

$$P = 2T/r$$

**POWER:** Surface Tension Trap.  
Without Surfactant Sid,  
Alfie collapses during exhale.

**LORE:** Small alveoli have higher pressure than large ones. No DPPC = NRDS.



**BLÖDSINN!**  
Check the  
PEEP!

# COMPLIANCE CONNIE (THE ELASTICITY QUEEN)

$$\Delta V / \Delta P$$

**POWER:** The Recoil Snap.  
Determines how easily  
the bellows expand.

High in Emphysema (floppy), Low in Fibrosis (stiff).

**LORE:** High in Emphysema (floppy), Low in Fibrosis (stiff).



# DEAD-SPACE DAN (THE GULP)

$$V_d = V_t \times \left( \frac{PaCO_2 - PeCO_2}{PaCO_2} \right)$$

**POWER:** The Empty Breath.  
Air that talks but never trades.  
Air that talks but never trades.

**LORE:** Anatomical (conducting zone) vs. Physiological (unperfused alveoli).



# SHUNT-MAN SHAUN (THE SHORT CIRCUIT)

$$V/Q = 0$$

**POWER:** Oxygen Sabotage.  
Perfusion without ventilation.

**LORE:** Blood bypasses the gas. Immune to 100% O<sub>2</sub> therapy.





# HEMOGLOBIN HANK (THE OXYGEN BUS)

## SIGMOID CURVE

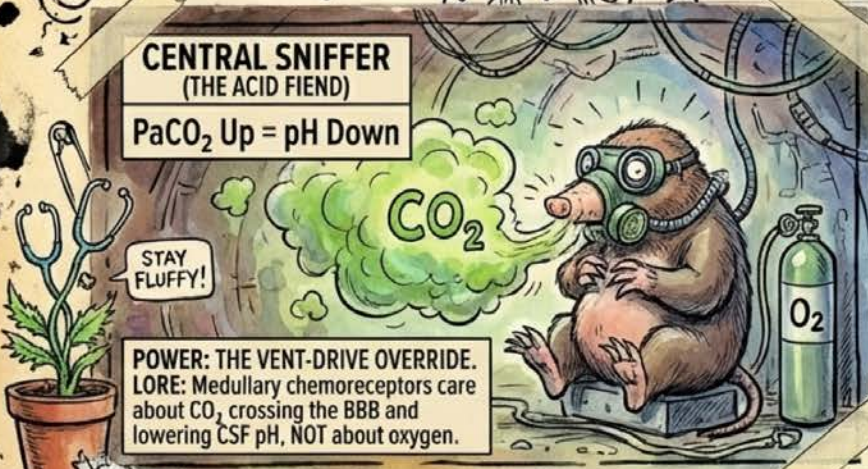


POWER: COOPERATIVE LOADING.

LORE: Four seats. Once one is taken, the rest are easier to fill.

# CENTRAL SNIFFER (THE ACID FIEND)

$\text{PaCO}_2$  Up = pH Down



POWER: THE VENT-DRIVE OVERRIDE.

LORE: Medullary chemoreceptors care about  $\text{CO}_2$  crossing the BBB and lowering CSF pH, NOT about oxygen.

LORE: Medullary ignores a pristine, untouched tank only.

# ZONAL ZACK (THE GRAVITY RULER)

$\text{P}_A > \text{P}_a > \text{P}_v$

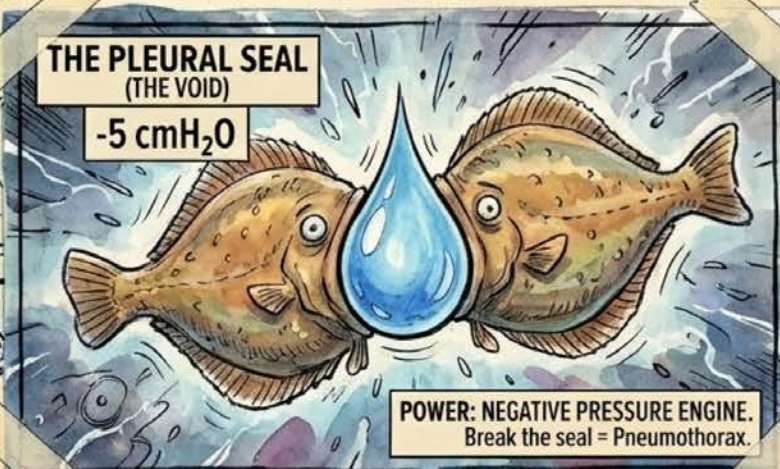
POWER: THE V/Q LADDER.

LORE: Ventilation and Perfusion both increase at the base, but Perfusion increases more.

LORE: High in Emphysema (floppy), Low in Fibrosis (stiff).

# THE PLEURAL SEAL (THE VOID)

$-5 \text{ cmH}_2\text{O}$



POWER: NEGATIVE PRESSURE ENGINE.

Break the seal = Pneumothorax.

LORE: Keeps lungs expanded and chest wall in. Break entirely.



## RIGHT-SHIFT RICK (THE EJECTOR SEAT)

CO<sub>2</sub> Up, Acid Up, Heat Up

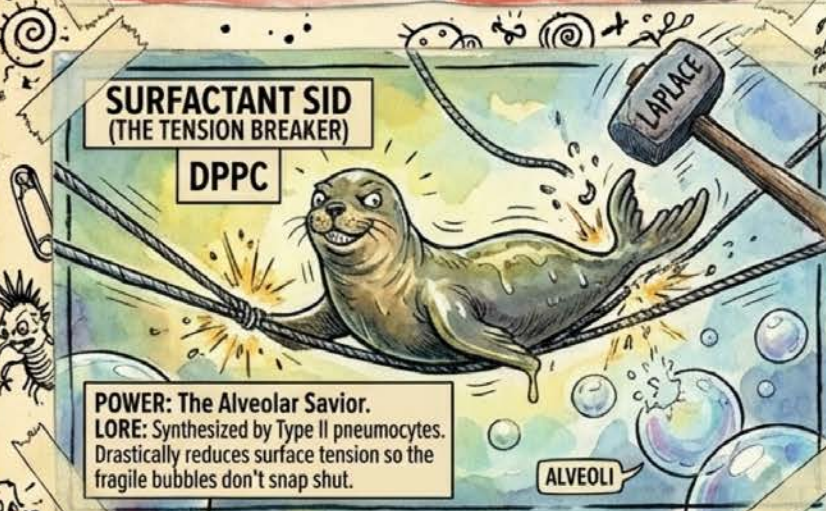


**POWER:** The Oxygen Dump.

**LORE:** Shifts the Sigmoid curve right. Tissues need oxygen during exercise, Rick forces Hemoglobin Hank to let it go.

## SURFACTANT SID (THE TENSION BREAKER)

DPPC



**POWER:** The Alveolar Savior.

**LORE:** Synthesized by Type II pneumocytes. Drastically reduces surface tension so the fragile bubbles don't snap shut.

ALVEOLI

**LORE:** Snapping the iron ron strings a:biatqii, untouched tank only.

## LEFT-SHIFT LARRY (THE DOOR LOCK)

Cold Up, pH Up



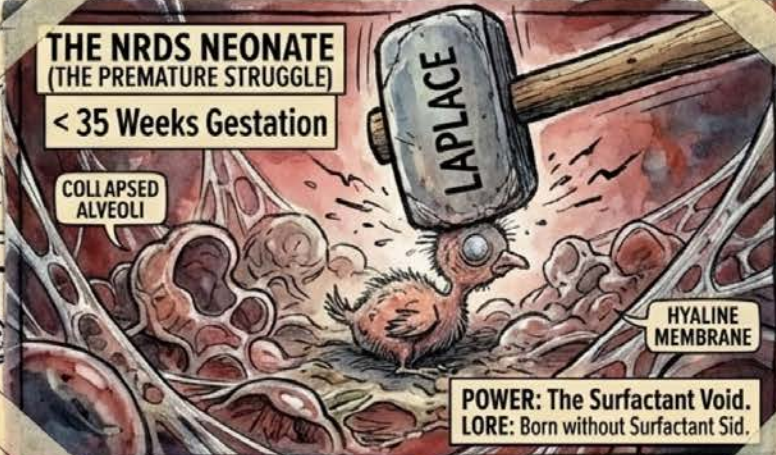
**POWER:** The Hoarder's Grip.

**LORE:** Shifts the curve left. Hemoglobin holds tighter to oxygen. O<sub>2</sub> cannot reach the starving tissues.

**LORE:** High in Emphysema (floppy), Low in Fibrosis (stiff).

## THE NRDS NEONATE (THE PREMATURE STRUGGLE)

< 35 Weeks Gestation



COLLAPSED  
ALVEOLI

HYALINE  
MEMBRANE

**POWER:** The Surfactant Void.  
**LORE:** Born without Surfactant Sid.

**LORE:** Born without Surfactant Sid. Alveoli collapse, causing massive intrapulmonary shunting and hyaline membrane formation.

GW

Georg v. Westphalen



**OBSTRUCTION OTIS**  
(THE GATEKEEPER)

**FEV1/FVC < 0.7**

**POWER:** AIR TRAPPING. HE LETS YOU IN, BUT HE WON'T LET YOU LEAVE.  
**LORE:** SCOOPED-OUT FLOW-VOLUME LOOP. OTIS LIVES IN THE SMALL AIRWAYS.



**EMMA EMPHYSEMA**  
(THE PINK PUFFER)

**DLCO DOWN**

**POWER:** THE PURSED-LIP PANIC.  
**LORE:** CENTRIACINAR (SMOKERS) VS PANACINAR (A1AT). DESTROYED SEPTA = LOST SURFACE AREA.



**LORE:** Snapping the iron ron strings arbiatgail, untouched tank only.

**BARRY BRONCHITIS**  
(THE BLUE BLOATER)

**REID INDEX > 50%**

**POWER:** MUCUS HYPERTROPHY.  
**LORE:** PRODUCTIVE COUGH FOR 3 MONTHS OVER 2 YEARS. LEADS TO COR PULMONALE.



**LORE:** High in Emphysema (floppy), Low in Fibrosis (stiff).

**ASTHMA ASH**  
(THE TWITCHY AIRWAY)

**REVERSIBLE BRONCHOCONSTRICTION**



**POWER:** THE CURSCHMANN SPIRAL.  
**LORE:** TYPE 1 HYPERSENSITIVITY. IgE-MEDIATED MAST CELL MAYHEM.

**LORE:** Type 1 Hypersensitivity. IgE-mediated mast cell mayhem.

**GW**

Georg v. Westphalen



# SABA Sam (The Rescue Sprint)

Beta-2 Agonist



POWER: The Instant Relief.  
LORE: Fast onset bronchodilation.  
Albuterol smooth muscle relaxation.

# LAMA Llama (The Shield)

M3 Antagonist



POWER: Drying the Swamp.  
LORE: The marathon runner. Blocks parasympathetic bronchoconstriction.

Diagnosis?!



# CF-Charlie (The Salt Trap)

Cl- Channel Defect



POWER: The Viscous Blockade.  
LORE: Delta F508 mutation.  $\text{Cl}^-$  stuck inside,  $\text{Na}^+$  and  $\text{H}_2\text{O}$  follow, leaving dehydrated, thick mucus in airways and pancreas.

# Theophylline Tiger (The PDE Predator)

Narrow Therapeutic Index

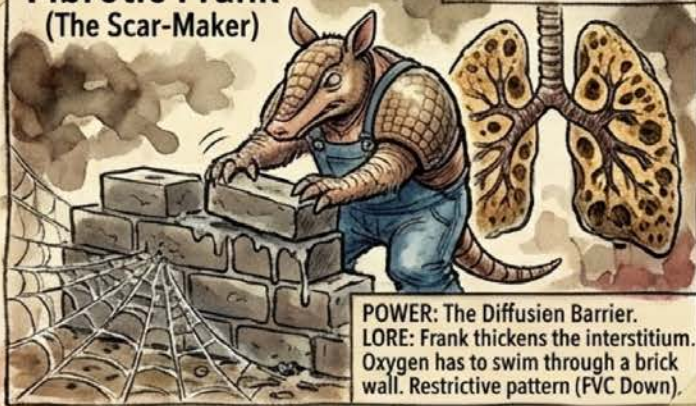


POWER: cAMP Accumulation.  
LORE: Phosphodiesterase inhibitor. Great bronchodilator, but easy to overdose (seizures, arrhythmias).



## Fibrotic Frank (The Scar-Maker)

Honeycombing on CT



**POWER:** The Diffusion Barrier.  
**LORE:** Frank thickens the interstitium. Oxygen has to swim through a brick wall. Restrictive pattern (PVC Down).

## Dusty Dan (The Overloaded Macrophage)

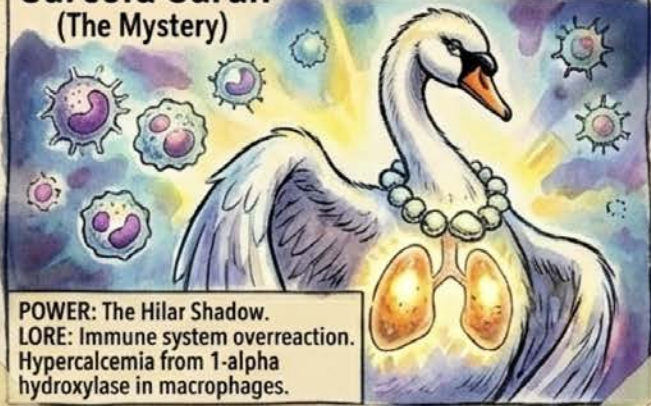
General Pneumoconiosis



**POWER:** The Fibrotic Trigger.  
**LORE:** Macrophages eat inorganic dust, die, and release fibrogenic cytokines.

## Sarcoid Sarah (The Mystery)

Non-caseating Granulomas / ACE Up



**POWER:** The Hilar Shadow.  
**LORE:** Immune system overreaction. Hypercalcemia from 1-alpha hydroxylase in macrophages.

## Asbestos Abby (The Plumber)

Ferruginous Bodies



**POWER:** The Mesothelioma Timebomb.  
**LORE:** From roofing/shipbuilding. Affects lower lobes. Bronchogenic carcinoma > Mesothelioma.

What the  
FLUFF?!



What the  
FLUFF?!





## Silica Sid (The Upper Lobe Sandblaster)

Eggshell Calcifications



**POWER:** The Macrophage Killer.  
**LORE:** Sand disrupts phagolysosomes.  
Massive risk of TB susceptibility.  
Lives in the upper lobes.

## Hypersensitivity Harry (The Farmer's Cough)

Restrictive Pattern



**POWER:** The Organic Shrink.  
**LORE:** Allergic response to organic dusts (bird fancier's, farmer's lung). Removing the trigger stops the fibrosis.

## Coal Carl (The Black Lung)

Anthraxosis



**POWER:** The Carbon Stiffening.  
**LORE:** Coal worker's pneumoconiosis.  
Massive progressive fibrosis.

## Obesity Oliver (The Heavy Blanket)

Extrapulmonary Restriction



**POWER:** The Upward Crush.  
**LORE:** The lungs are healthy, but the bellows cannot open.  
A classic cause of extrapulmonary restrictive defect.





**Rusty the Strep** (*S. pneumoniae*)



**Gram+ Diplococci**

**POWER:** Lobar Consolidation  
**LORE:** The King of Community Acquired Pneumonia. Rusty sputum is his calling card.



**The TB Tank** (*Mycobacterium*)



**Acid-Fast / Caseating Granuloma**

**POWER:** The Apical Squatter  
**LORE:** Loves Oxygen. Ghon Complexes are his initial bunkers. Reactivation happens at the top.

**Walking Wally** (*Mycoplasma*)



**No Cell Wall / Cold Agglutinins**

**POWER:** The Mime Attack  
**LORE:** Atypical pneumonia. Ghost in the machine. Erythema multiforme association.



**Klebsiella Kelly** (*The Jelly*)



**Bulging Fissure Sign**

**POWER:** The Alcoholic Aspiration  
**LORE:** Enteric flora aspirated into the lungs. Necrofizing lobar pneumonia.



**HUMBUG!**



# PJP Paul (The Opportunist)

Silver Stain / CD4 < 200



POWER: The Immunocompromised Strike.  
LORE: *Pneumocystis jirovecii*.  
Bilateral ground-glass opacities.  
Treat with TMP-SMX.

# The Abscess Axe (The Anaerobe Cleaver)

Air-Fluid Level



POWER: The Foul Breath.  
LORE: Aspiration of oropharyngeal flora. *Peptostreptococcus*, *Fusobacterium*. Treat with Clindamycin.

# Legionnaire Leo (The Water Tower)

Silver Stain / Hyponatremia



POWER: The Aerosol Ambush.  
LORE: Atypical pneumonia with  
GI/CNS symptoms and low sodium.  
Needs a urine antigen test.

# Viral Vicky (The Interstitial Ghost)

Atypical Interstitial Infiltrate



POWER: The Ground-Glass Glitch.  
LORE: Causes atypical pneumonia. Leaves the  
alveolar space empty but inflames the walls.

# KayfaBIZARRO!



Look at  
my spiral!

My flagella  
are fabulous!

Cocci with  
the mostest!

GW

Georg v. Westphalen



PUNK LUNGS FOREVER!

WATCH OUT!

## Rifampin Rick (The Orange Painter)

RNA Polymerase Inhibitor



POWER: The CYP450 Rev-Up.

LORE: Induces liver enzymes. Warn the patient about the orange fluids!

## Isoniazid Izzy (The Nerve Slicer)

Mycolic Acid Inhibitor



POWER: The Wall Breaker.

LORE: Causes peripheral neuropathy. Always administer with Vitamin B6 (Pyridoxine).

## Pyrazinamide Pyro (The Gout Bringer)

Uric Acid Up



POWER: The Acidic Ambush.

LORE: Kills TB in the acidic phagolysosome. Hyperuricemia is the infamous side effect.

## Ethambutol Ed (The Color-Blind Sniper)

Arabinosyltransferase Inhibitor



POWER: The Wall Blocker.

LORE: Causes optic neuropathy. Red-green color blindness. Eye-thambutol.



## Benny the Blockage

(Pulmonary Embolism)

S1Q3T3 / V/Q = Infinity



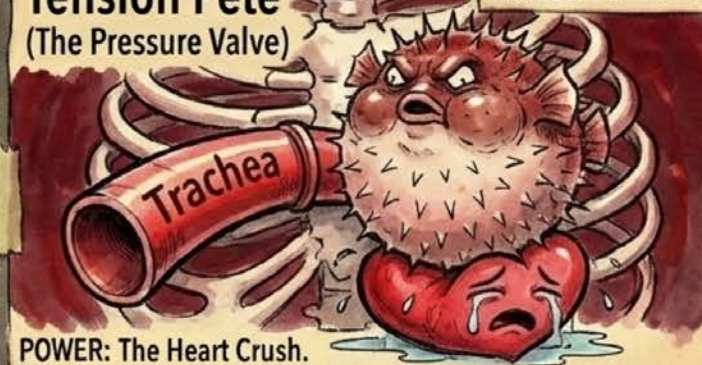
POWER: The Sudden Gasp.

LORE: Virchow's Triad survivor. Stops the flow, but air keeps coming. Dead Space Dan's master.

## Tension Pete

(The Pressure Valve)

Tracheal Deviation



POWER: The Heart Crush.

LORE: One-way valve in the pleura. Every breath makes Pete bigger, dropping venous return to zero.

## Small-Cell Sam

(The Neuroendocrine Ninja)

Central Location / ACTH / ADH



POWER: Paraneoplastic Spells.

LORE: Smoker's cancer. Doesn't just grow; secretes chaos. Inoperable, treat with chemo.

## Squamous Sid

(The Smoker's Squat)

PTHrP / Keratin Pearls



POWER: The Calcium Spike.

LORE: Central, cavitary, cigarette-related. Ectopic parathyroid hormone-related peptide.

Stay fluffy!

KayfaBINGO!

Stay fluffy!

OR  
BUST!



**PULM BOSS BATTLE!**

## The Pulm Attending (The Synthesis Boss)

A-a Gradient = (Age/4) + 4

ABG:  
pH 7.32  
PaCO<sub>2</sub> 55  
PaO<sub>2</sub> 60  
HCO<sub>3</sub> 28

A-a GRADIENT: **ELEVATED!**

POWER: The Ultimate Pimp.

LORE: Is it Hypoventilation, V/Q Mismatch, Shunt, or Diffusion? Explain the gradient or get out of my ICU.



## Alpha-1 AI (The Double Threat)

Panacinar Emphysema / Cirrhosis



POWER: The Elastase Rot.

LORE: Lack of alpha-1 antitrypsin leaves elastase uninhibited. Destroys the lower lobes.

## The Cor Pulmonale King (The Hypoxic Backup)

(Right Ventricular Hypertrophy)

**PULMONARY  
HYPERTENSION**

**SNAP!**

POWER: The Right-Sided Break.

LORE: Chronic hypoxia causes pulmonary vasoconstriction, breaking the right heart.



## Berty Bronchiectasis (The Dead Pipes)

Dilated Airways



POWER: The Permanent Stretch.

LORE: Chronic necrotizing infections destroy airway muscle and elastic tissue.

BLÖDSINN!

KayfaBINGO!

HYPOXIA  
KILLS!

PUNK  
PULM!

PUNK PULM!

GW

Georg v. Westphalen



QUEST:

# The Blue Flight

AaDO<sub>2</sub> Up Up, PaCO<sub>2</sub> Down



POWER:

The D-Dimer Trap.

LORE: 70yo smoker, 12-hour flight, sudden SOB, clear X-ray, it's Benny. The A-a gradient is sky-high due to V/Q mismatch.



QUEST:

# The Night Shift Shadow

Stat: Apical Cavitation



POWER: The Cytokine Burn.

LORE: 3-week cough, night sweats, weight loss. The Tank is reactivating in the oxygen-rich apex. Deploy the RIPE Squad.

STAY FLUFFY!

GET RIPE!

ISOLATE!

QUEST:

# The Salty Kiss

Sweat Chloride > 60

SALT!  
SALT!  
SALT!



POWER: The Pancreatic Block.

LORE: Toddler, recurrent pneumonia, foul stools. Charlie trapped the chloride, blocking pancreatic enzymes.



QUEST:

# The Silent Choke

Stat: Normal A-a Gradient,  
High CO<sub>2</sub>



HYPOXIA KILLS!



POWER: The Drive Drop.

LORE: Hypoventilation. The lungs work fine (A-a is normal), but the brain forgot to breathe.

WAKE UP!

PUNK PULM!

BLÖDSINN!

NO DRIVE!



**Hypoxemia ( $\text{PaO}_2 < 80$ ) → Check the A-a Gradient!**

**Normal A-a Gradient**  
(The Bellows stalled!)

**High  $\text{CO}_2$ ?**

**Hypoventilation**  
(Central Sniffer asleep!  
Opioids, ALS).

**Normal  $\text{CO}_2$ ?**

**High Altitude**  
(Not enough  $\text{O}_2$  in the arena!).

Diagnosis?!  
**BLÖDSINN!**

**Elevated A-a Gradient**  
(The Lungs are damaged!)  
→ Give 100% Oxygen!

**Corrects with  $\text{O}_2$ ?**

**V/Q Mismatch**  
(Benny the Blockage,  
Obstruction Otis, Fibrotic Frank).  
Airway or blood is clogged,  
but sheer force of  $\text{O}_2$   
overcomes it.

**DOES NOT  
Correct with  $\text{O}_2$ ?**

**SHUNT!**  
(Shunt-Man Shaun).  
The blood completely  
bypassed the alveoli.  
 $\text{O}_2$  therapy does nothing!